FIFTH LESSON

Grains of Space

Despite certain obscurities, infelicities and still unanswered questions, the physics I have outlined provides a better description of the world than we have ever had in the past. So we should be quite satisfied. But we are not.

There's a paradox at the heart of our understanding of the physical world. The twentieth century has given us the two gems of which I have spoken: general relativity and quantum mechanics. From the first cosmology developed, as well as astrophysics, the study of gravitational waves, of black holes, and much else besides. The second provided the foundation for atomic physics, nuclear physics, the physics of elementary particles, the physics of condensed matter, and much, much more. Two theories, profligate in their gifts, which are fundamental to today's technology and have transformed the way we live. And yet the two theories cannot both be right, at least in their current form, because they contradict each other.

A university student attending lectures on general relativity in the morning and others on quantum mechanics in the afternoon might be forgiven for concluding that his professors are fools, or have neglected to communicate with each other for at least a century. In the morning the world is curved space where everything is continuous; in the afternoon it is a flat space where quanta of energy leap.

The paradox is that both theories work remarkably well. Nature is behaving with us like that elderly rabbi to whom two men went in order to settle a dispute. Having listened to the first, the rabbi says: 'You are in the right.' The second insists on being heard, the rabbi listens to him and says: 'You're also right.' Having overheard from the next room the rabbi's wife then calls out, 'But they can't *both* be in the right!' The rabbi reflects and nods before concluding: 'And you're right too.'

A group of theoretical physicists scattered across the five continents is laboriously trying to settle the issue. Their field of study is called 'quantum gravity': its objective to find a theory, that is, a set of equations – but above